AI-NIDS: AI-Based Real-Time Network Intrusion Detection System

Comprehensive System Architecture & Technical Project Manual

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1. Executive Summary & Project Overview

AI-NIDS (Artificial Intelligence Network Intrusion Detection System) is a high-performance, real-time cybersecurity platform designed to ingest raw network packet flows, extract statistical and behavioral flow features, classify anomalous traffic using machine learning inference models, and present authoritative telemetry across a centralized Security Operations Center (SOC) dashboard.

The system is built to operate in both single-machine demonstration modes and a physical two-PC laboratory cyber range, strictly separating the Remote Attack Generation & Command-and-Control (C2) plane from the Sensor Capture, AI Inference, and SOC Detection Authority plane.

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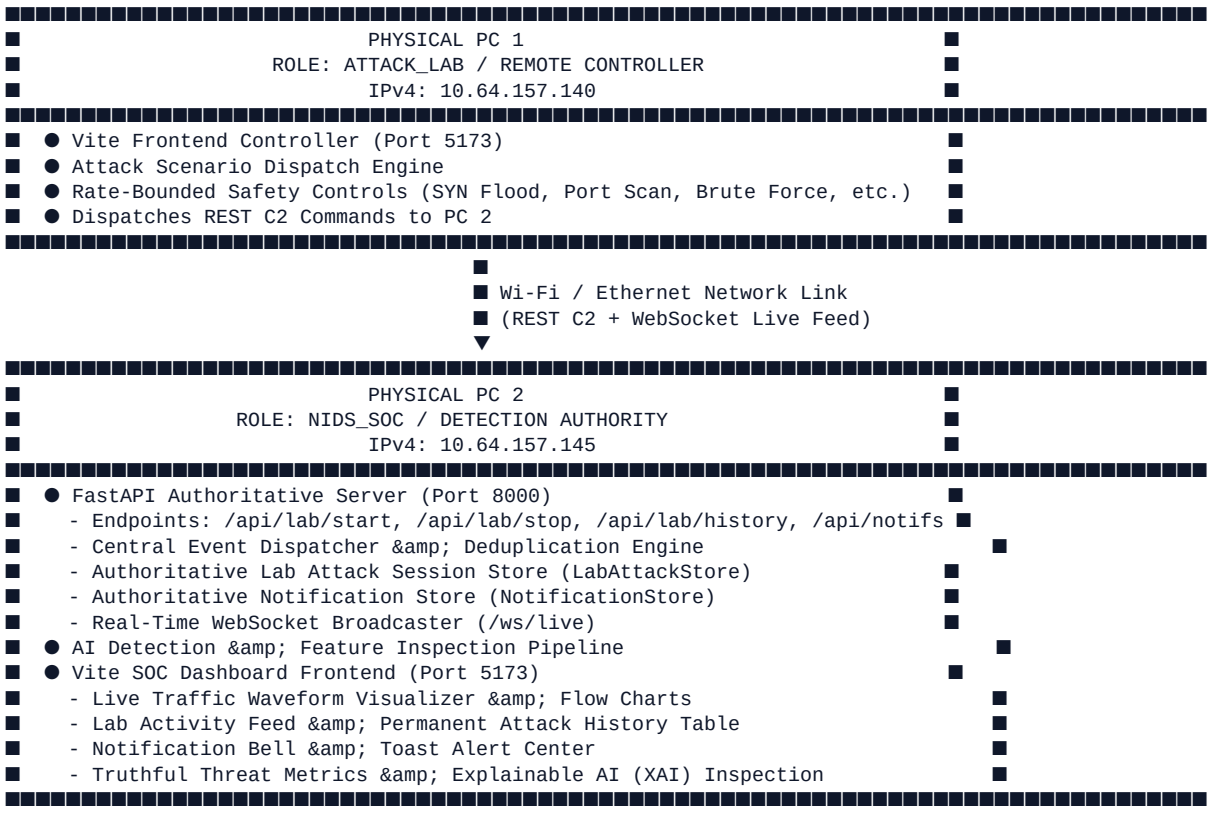
2. Core Architectural Roles (Two-PC Cyber Range)



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3. High-Level System Workflow & Data Flow

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sequenceDiagram
    autonumber
    participant PC1 as PC 1: Attack Lab (10.64.157.140)
    participant API as PC 2: FastAPI Backend (10.64.157.145:8000)
    participant Store as PC 2: LabAttackStore &amp;amp; NotifStore
    participant WS as PC 2: WebSocket Broadcaster (/ws/live)
    participant SOC as PC 2: SOC Dashboard UI

    PC1-&gt;&gt;API: POST /api/lab/start { scenario: "DOS_SYN_FLOOD", source_ip: "10.64.157.140", target_ip: "10.64.157.145" }
    API-&gt;&gt;Store: Generate attack_id ("lab_attack_..."), insert ACTIVE record
    API-&gt;&gt;Store: Create LAB_ATTACK_STARTED notification
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API-&gt;&gt;WS: Broadcast ATTACK_STARTED event payload
WS--&gt;&gt;SOC: Deliver ATTACK_STARTED message
SOC-&gt;&gt;SOC: Display Toast Alert + Increment Bell counter + Update Active Scenario

Note over PC1,PC2: Scenario Transmits for Bounded Duration (e.g. 20s)

API-&gt;&gt;Store: Update attack_id status -&gt; "COMPLETED", record stopped_at
API-&gt;&gt;Store: Create LAB_ATTACK_STOPPED notification
API-&gt;&gt;WS: Broadcast ATTACK_STOPPED event payload
WS--&gt;&gt;SOC: Deliver ATTACK_STOPPED message
SOC-&gt;&gt;SOC: Display Completion Toast + Transition Active Banner to STANDBY
SOC-&gt;&gt;SOC: Permanently retain session in RECENT LAB ACTIVITY table
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4. Key Subsystems & Modules

4.1. Network Sensor & Telemetry Ingestion

- Ingests packet flows across network interfaces (Wi-Fi / Ethernet / Loopback).
- Computes real-time bandwidth (bytes/sec), packet throughput (packets/sec), active bidirectional flow counts, and protocol distributions (TCP, UDP, ICMP).
- Provides dynamic network interface enumeration and diagnostic health checks.

4.2. Feature Extraction Pipeline

Extracts standard statistical flow features compatible with benchmark intrusion datasets (such as CICIDS2017 and UNSW-NB15):

- Flow Duration & Inter-Arrival Times (IAT): Mean, standard deviation, and maximum delta between packets.
- Flag Distributions: SYN, ACK, FIN, RST, PSH, and URG packet ratios.
- Directional Volume Asymmetry: Forward packets/bytes vs. Backward packets/bytes.
- Shannon Entropy: Evaluates source IP entropy and destination port entropy to catch distributed scanning and DDoS activity.

4.3. AI Inference & Explainable AI (XAI) Engine

- Evaluates extracted flow features against gradient-boosted decision tree algorithms (XGBoost) and anomaly models (Isolation Forest).
- Produces normalized anomaly scores (0.0 to 1.0), risk scores (0 to 100), and classification probabilities.
- Explainability Factors (XAI): Generates weighted top-contributing factors explaining why a particular flow was flagged (e.g., SYN Packet Ratio: 99.1% [EXCESSIVE], Flow Duration: 12ms [ABNORMAL]).

4.4. Authoritative Event Store & Notification Pipeline

- Central Event Dispatcher: All live events (ATTACK_STARTED, ATTACK_STOPPED, THREAT_DETECTED, THREAT_RESOLVED) are validated, given unique event IDs (evt_...), and recorded in a memory-bounded LIFO ring buffer.
- LabAttackStore: Retains completed and active lab simulation sessions indexed by authoritative attack_id.
- NotificationStore: Maintains notification items (notif_...) with read/unread tracking, timestamp, and metadata.
- REST Endpoints:
- GET /api/lab/history: Returns full persisted lab attack history.
- GET /api/notifications: Returns full notification history.
- POST /api/notifications/read/{id}: Marks an individual alert as read.
- POST /api/notifications/read-all: Marks all notifications as read.
- Page Refresh Resilience: The frontend automatically reconciles state on startup using REST endpoints before listening for live WebSocket messages.

4.5. SOC Visual Dashboard & User Interface

- Truthful Metric Separation: Clearly isolates real security detections (Threats Detected: 0) from controlled lab events (Lab Attacks: N, Active Lab Attacks: 0/1, Control Events: N).
- Real-Time Traffic Charting: Interactive area charts displaying live packets/sec and flow metrics with selectable time ranges (1M, 5M, 15M, 1H).
- Interactive Network Topology: Visual diagram showing PC 1 Attacker, Communication Link, and PC 2 NIDS/Victim nodes.
- Top Talkers & Connection Flow Table: Detailed inspection of active socket pairs with protocol, bandwidth, and anomaly indicators.
- Notification Bell Dropdown: Topbar dropdown showing unread badge counts, alert items, and mark-as-read controls.

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5. Supported Attack Simulation Scenarios

Scenario ID	Name	Category	Protocol	Default Intensity & Window	Description
PORT_SCAN	SYN Stealth Recon Scan	Reconnaissance	TCP	MEDIUM (15s)	Rapid SYN packets across multiple destination ports (1-1024) to identify open listener services.
BRUTE_FORCE	SSH / HTTP Credential Attack	Credential Attacks	TCP	MEDIUM (18s)	High-frequency connection churn and authentication handshakes against port 22 / 8080.
DOS_SYN_FLOOD	TCP SYN Flood (Single Source)	Network Availability	TCP	HIGH (20s)	High-rate SYN packet flood with zero ACK completion to saturate listener connection tables.
DDOS_DISTRIBUTED_FLOOD	Distributed SYN Flood	Network Availability	TCP	HIGH (20s)	Simulated botnet cluster multi-socket availability attack demonstrating source IP entropy surge.
DNS_EXFILTRATION	DNS Tunneling & Data Exfiltration	Protocol Anomaly	UDP	MEDIUM (15s)	Encoded base64/hex payloads hidden inside anomalous subdomain TXT query requests on port 53.
SUSPICIOUS_FLOW	Low-and-Slow C2 Beacon	Behavioral Anomaly	TCP	LOW (15s)	Periodic heartbeat beaconing with fixed inter-arrival times and non-standard TLS headers.

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6. Safety Controls & Cyber Range Governance

• Target Lock Restriction: Simulation dispatch requests are strictly locked to designated laboratory targets (e.g. 10.64.157.145). Dispatches to unauthorized addresses are rejected with HTTP 403.

• Hard Bounded Timers: Every simulation scenario has an internal timeout (10s to 20s) and automatically triggers an authoritative ATTACK_STOPPED signal upon expiration.

• Identity Contract Enforcement: In two-PC mode, the system guarantees that source_ip (10.64.157.140) is never overwritten by the NIDS server's host (10.64.157.145).